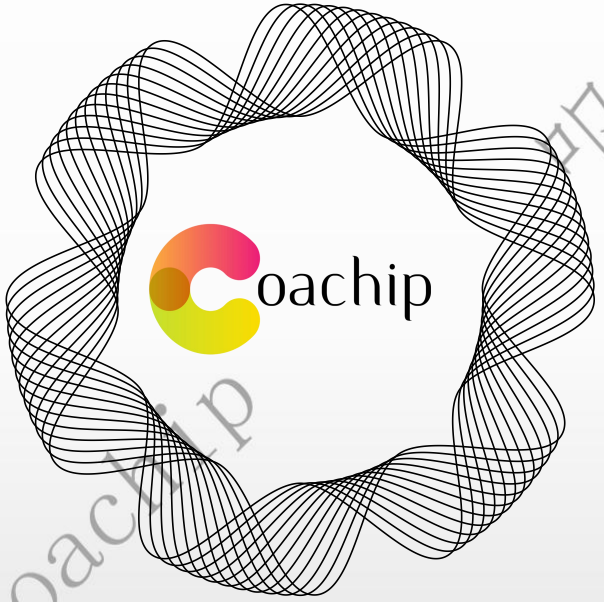



# SV随机约束常见错误

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# 课程内容及目标

## 课程内容：

- 了解并避免一些随机化约束的错误




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# 有符号变量导致的随机化错误

```
class SignedVars;
  rand byte pkt1_len, pkt2_len; // signed type
  constraint total_len
  {
    pkt1_len + pkt2_len == 64;
  }
endclass
```

Obviously, you could get pairs of values such as (32, 32) and (2, 62).

Additionally, you could see (–63, 127), as this is a legitimate solution of the equation.

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# 变量溢出导致的随机化错误

```
class Vars32;
   rand bit [31:0] pkt1_len, pk2_len; //unsigned type
    constraint total_len
  {
   pkt1_len + pk2_len == 64;
  }
endclass
```

- Even this version causes problems, as large values of pkt1_len and pkt2_len, such as 32'h80000040 and 32'h80000000

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# System Verilog随机化的常见错误

改正：

```
class Vars8;
  rand bit [7:0] pkt1_len, pkt2_len; // 8-bits wide
  constraint total_len
  {
    pkt1_len + pkt2_len == 9'b64; // 9-bit sum
  }
endclass
```

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# 避免随机化错误的一些技巧：

- 避免使用乘除和取模的操作
- 如果非要使用，用左移或者右移来代替乘除
- 用AND mask来代替取模的操作；

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# Q&A

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